

Free Cash Flow Practice Problem Solutions

Professor Gormley

These two problems will help refresh your memory on calculating FCFs and how financial statements are linked together.

1. Consider the following financial statements:

Year-end balance sheet		
Year	0	1
Assets		
Cash	100	90
Other Current Assets	150	153
Fixed assets		
Gross Fixed Assets	1,000	1,150
Accumulated Deprec.	<u>(300)</u>	<u>(415)</u>
Total	950	978
Liabilities		
Current Liabilities	70	78
Debt (short & long term)	380	337
Equity		
Stock	400	400
Retained Earnings	<u>100</u>	<u>163</u>
Total	500	563
Profit & Loss		
Sales	1,000	1,050
COGS (excluding depreciation)	(700)	(730)
Depreciation (annual expense)	(100)	(115)
Interest	<u>(30)</u>	<u>(30)</u>
Profit before Tax	170	175
Provision for Income Taxes	<u>(68)</u>	<u>(70)</u>
Profit after Tax	102	105
Dividend	<u>(40)</u>	<u>(42)</u>
Retained Earnings	62	63

- a. What is the earnings before interest and taxes (EBIT) in Year 1?

$$\text{EBIT} = \text{Profit before tax} + \text{Interest Expense}$$

$$\text{EBIT} = 175 + 30 = 205$$

- b. If you assume the tax rate for this firm is 40%, what was this firm's operating cash flow (OCF) in Year 1? *[Remember, OCF is just another term used for unlevered net income + depreciation]*

Remember, OCF is just another term used by some to describe unlevered net income + depreciation... I.e.,

$$\text{OCF} = \text{EBIT} * (1 - \text{Tax Rate}) + \text{Depreciation}$$

$$\text{OCF} = 205 * (1 - 0.40) + 115 = 238$$

- c. What as the capital expenditures (CapEx) of the firm in Year 1?

Two possible ways to get this...

$$\text{CapEx} = \text{End gross fixed assets} - \text{Start gross fixed assets}$$

$$\text{CapEx} = 1150 - 1000 = 150$$

or

$$\text{CapEx} = \text{End net fixed assets} - \text{Start net fixed assets} + \text{Depr.}$$

$$\text{CapEx} = (1150 - 415) - (1000 - 300) + 115 = 150$$

- d. What is the change in net working capital from Year 0 to Year 1?

$$\text{NWC} = \text{Current assets} - \text{Current Liabilities}$$

Which, for this firm equals...

$$\text{NWC} = (\text{Cash} + \text{Other Current Assets}) - \text{Current Liabilities}$$

$$\text{NWC in year 0} = (100 + 150) - 70 = 180$$

$$\text{NWC in year 1} = (90 + 153) - 78 = 165$$

$$\text{Change in NWC} = -15$$

- e. Finally, what was the FCF in Year 1?

$$\text{FCF} = \text{OCF} - \text{NCS} - \Delta \text{ in NWC}$$

$$\text{FCF} = 238 - 150 - (-15) = 103$$

2. Suppose the CEO of Bayer has decided it's time to make plans for next year. To do this, he has asked his financial team to construct a projection of what they think the balance sheet and income statement will look like for next year (i.e., year $T+1$). These projected statements, which we call a pro forma statement, and the current year's financial statement (i.e., year T) are provided below. Using this, calculate the projected FCF of Bayer for next year.

$$\begin{aligned}\text{OCF} &= \text{EBIT} + \text{Depreciation} - \text{Tax Rate} * \text{EBIT} \\ \text{OCF} &= (10056) + (5000) - (0.40 * 10056) = 11033.6\end{aligned}$$

$$\begin{aligned}\text{CapEx} &= \text{NFA}_{T+1} - \text{NFA}_T + \text{Depreciation} \\ \text{CapEx} &= (30250) - (25000) + 5000 = 10250\end{aligned}$$

$$\begin{aligned}\text{NWC in 2007} &= 20,546 - 1,356 = 19,190 \\ \text{NWC in 2008} &= 22,412 - 5,956 = 16,456 \\ \text{Change in NWC} &= -2,734\end{aligned}$$

$$\begin{aligned}\text{FCF} &= \text{OCF} - \text{CapEx} - \text{Change in NWC} = \\ \text{FCF} &= 11033.6 - 10250 + 2,734 = 3,517.6\end{aligned}$$

Financial Statements for Bayer

BALANCE SHEET -----	Year T+1	Year T
ASSETS		
Cash and Cash Equivalents	1,424	2,067
Account Receivables	2,985	2,376
Inventory	18,003	16,103
Total Current Assets	22,412	20,546
Net Fixed Assets	30,250	25,000
Total Assets	52,662	45,546
LIABILITIES		
Accounts payable	1,392	567
Taxes Payable	4,564	789
Total Current Liabilities	5,956	1,356
Long term debt	22,725	24,259
Total Liabilities	28,681	25,615
Total Equity	23,981	19,931
Common Stock	15,000	15,000
Retained Earnings	8,981	4,931
Total Equity and Debt	52,662	45,546
INCOME STATEMENT -----	Year T+1	
Sales	59,595	
Costs (includes COGs, SG&A, and Depreciation)	49,539	
EBIT	10,056	
Interest Expenses	1,056	
Taxes (40%)	3,600	
Net Income	5,400	
Dividends	1350	
Retained Earnings	4050	
Depreciation	5000	